

Thomson Problem for $N = 8$: Overview

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1 Introduction

For N distinct points $x_1, \dots, x_N$ on the unit sphere $S^2 \subset \mathbb{R}^3$, the *Thomson problem* asks for the configuration that minimises the Coulomb energy

$$E(x) = \sum_{i < j} \frac{1}{\|x_i - x_j\|}. \quad (1)$$

The problem goes back to Thomson [8]. Rigorous solutions are known for only a few values of N .

For $N = 8$, the square antiprism has long been the conjectured minimiser, proved rigorously by Kryvonos, Liehr, and Taylor [5]. We give a complete, machine-checked warrant of this fact in Lean 4, developed with the assistance of generative AI. This document gives an overview of the techniques used in that warrant.

2 Method

2.1 Statement: chords, inner products, and the energy

Definition 2.1 (Admissible). A tuple $x = (x_1, \dots, x_8) \in (S^2)^8$ is **admissible** if $\|x_i\| = 1$ for every i , and $x_i \neq x_j$ whenever $i \neq j$.

For an admissible x , write

$$t_{ij} := \langle x_i, x_j \rangle, \quad s_{ij} := \|x_i - x_j\| = \sqrt{2 - 2t_{ij}}, \quad (2)$$

for the inner product and *chord length* of the pair (i, j) , so that

$$E(x) = \sum_{i < j} \frac{1}{\sqrt{2 - 2t_{ij}}} = \sum_{i < j} \frac{1}{s_{ij}}. \quad (3)$$

2.2 The optimal antiprism height

Two squares of radius $r = \sqrt{1 - h^2}$ at heights $\pm h$, offset by 45° , form the *antiprism* $\text{antiprism}(h)$. Writing $u = h^2$, $\alpha = 2 - \sqrt{2}$, $\beta = 2 + \sqrt{2}$, its energy has the closed form (abusing notation, we write $E(u)$ for $E(\text{antiprism}(\sqrt{u}))$)

$$E(u) = \frac{4\sqrt{2} + 2}{\sqrt{1 - u}} + \frac{8}{\sqrt{\alpha + \beta u}} + \frac{8}{\sqrt{\beta + \alpha u}}. \quad (4)$$

Write u^* for the minimiser of E over this family:

$$u^* = 0.3140893679\dots, \quad E(u^*) = 19.6752878612\dots \quad (5)$$

2.3 The four chord types and five triangle types

At the optimal antiprism, the $\binom{8}{2} = 28$ pairwise chords take only four distinct values:

label	geometric meaning	count	chord length	inner product
A	side of a square	8	1.17125	$t_A = 0.31409$
B	near cross-square pair	8	1.28769	$t_B = 0.17092$
C	diagonal of a square	4	1.65639	$t_C = -0.37182$
D	far cross-square pair	8	1.89689	$t_D = -0.79910$

Correspondingly, of the $\binom{8}{3} = 56$ triangles $\{x_i, x_j, x_l\}$, only five combinatorial types occur, written as their three chord labels in decreasing order of length ($D > C > B > A$), occurring with multiplicities

$$DDA \ (8), \quad DCB \ (16), \quad DBA \ (16), \quad CAA \ (8), \quad BBA \ (8). \quad (6)$$

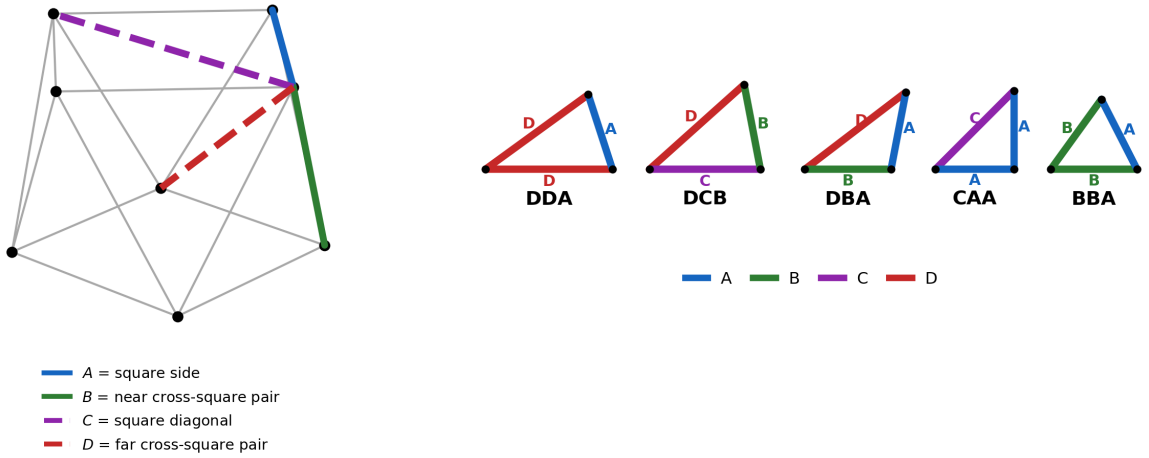


Figure 1: The antiprism, with one example edge of each chord type highlighted and the five triangle types, drawn to scale, with sides coloured by chord type.

2.4 Bounding the chords: linear programming and the separation theorem

The first step of the warrant is to put a *lower* bound on each pairwise chord length of a global minimum. Since $E(u^*) < 19.6753$, a global minimum x satisfies $E(x) \leq E(u^*) < 19.6753$. Suppose we have a function y of one chord length and a constant b such that, for every admissible x ,

$$y(s_{ij}) \leq \frac{1}{s_{ij}} \quad \text{for all } i \neq j, \quad \sum_{i < j} y(s_{ij}) = b + \rho(x) \quad \text{with } \rho(x) \geq 0. \quad (7)$$

Adding and subtracting $y(s_{ij})$ in each term of the energy gives the identity

$$E(x) = b + \sum_{i < j} \left(\frac{1}{s_{ij}} - y(s_{ij}) \right) + \rho(x), \quad (8)$$

in which every term after b is nonnegative. Discarding all of them gives the linear-programming bound $E(x) \geq b$; discarding all but one instead bounds the *slack* of a single pair by the excess of the energy over b ,

$$\frac{1}{s_{ij}} - y(s_{ij}) \leq E(x) - b \quad \text{for all } i \neq j. \quad (9)$$

For a global minimum, (9) gives $1/s_{ij} - y(s_{ij}) \leq \varepsilon$ with $\varepsilon := 19.6753 - b$. It is therefore enough to exhibit an s_0 with $1/s - y(s) > \varepsilon$ for every $0 < s \leq s_0$: no chord of a global minimum can then be at most s_0 . Multiplying by $s > 0$, that is the positivity on $(0, s_0]$ of the one-variable polynomial

$$P(s) = 1 - sy(s) - \varepsilon s, \quad (10)$$

and positivity is certified by the positivity of all coefficients of P in the Bernstein basis of $[0, s_0]$. Taking for y Yudin's polynomial [9] of degree 12 in t , which touches $1/s$ at three chord lengths and has $b = 19.6475639$, hence $\varepsilon = 0.0277361$, the polynomial P has degree 25 and all 26 of its Bernstein coefficients on $[0, 0.962]$ are positive. This two-point linear-programming argument [4, 9], applied to the near-optimal region, is the **separation theorem**:

$$E(x) \leq 19.6753 \implies s_{ij} > 0.962 \quad \text{for all } i \neq j, \quad (11)$$

equivalently $t_{ij} < 0.53728$. Below, these are used in the slightly weaker, rounder forms $t_{ij} \leq 0.5373$ and $s_{ij} \in [0.9619, 2]$. Any configuration that could match or beat the antiprism therefore has all of its chords in this range.

2.5 Defining the data structure

The Bachoc–Vallentin matrices. Bachoc and Vallentin [2] introduced three-point semidefinite bounds for spherical codes, built from a matrix kernel $S_k^n(u, v, t)$, symmetrised from a Chebyshev/Gegenbauer kernel so that $\sum S_k^n \succeq 0$ automatically over any finite point set on S^{n-1} ; Cohn and Woo [3] adapted these bounds to energy minimisation. For S^2 , truncated at degree d ($k \leq d$ and $i, j \leq d - k$; here $d = 8$, giving $(9 - k) \times (9 - k)$ blocks), with T_k the degree- k Chebyshev polynomial (here u, v, t denote generic inner products, unrelated to the antiprism parameter $u = h^2$):

$$Y_k(u, v, t)_{ij} = u^i v^j ((1 - u^2)(1 - v^2))^{k/2} T_k\left(\frac{t - uv}{\sqrt{(1 - u^2)(1 - v^2)}}\right), \quad (12)$$

$$S_k(u, v, t) = \frac{1}{6} \sum_{\sigma \in S_3} Y_k(u, v, t) \circ \sigma. \quad (13)$$

The classical two-point method of Delsarte, Goethals and Seidel [4], in Yudin's energy version [9], used in §2.4, bounds each pair separately, and for $N = 8$ this is not enough: Yudin's degree-12 polynomial gives $b = 19.6475639$, short of $E(u^*) = 19.675288 \dots$ by 0.0277, and no choice of y does better than about 19.6478. The three-point method below applies the same argument to triangles as well as pairs.

The energy can be written in two ways. As a sum over pairs, by definition,

$$E(x) = \sum_{i < j} \frac{1}{s_{ij}}, \quad (14)$$

and, since every pair $\{i, j\}$ lies in exactly 6 of the 56 triangles, as a sum over triangles,

$$E(x) = \frac{1}{6} \sum_{i < j < l} \left(\frac{1}{s_{ij}} + \frac{1}{s_{il}} + \frac{1}{s_{jl}} \right). \quad (15)$$

Taking $1 - 18\lambda$ times (14) plus 18λ times (15) splits the energy between the two, for any real λ :

$$E(x) = \underbrace{\sum_{i < j} (1 - 18\lambda) \frac{1}{s_{ij}}}_{\text{a sum over pairs}} + 3\lambda \underbrace{\sum_{i < j < l} \left(\frac{1}{s_{ij}} + \frac{1}{s_{il}} + \frac{1}{s_{jl}} \right)}_{\text{a sum over triangles}}. \quad (16)$$

The two sums are bounded by different methods, so the bound obtained depends on λ , which is a free parameter of the certificate. At $\lambda = 0$ the triangle sum drops out and the classical sum over pairs remains.

Bounding each sum. Each of the two sums is treated the way the classical method treats a sum over pairs: from each term, subtract a function of that term alone, chosen so that every difference (a *slack*) is nonnegative and the sum of the subtracted functions is bounded below by a constant independent of the configuration.

For the pairs, subtract a function L of the pair's inner product. It must lie below the pair's share, and $\sum_{i < j} L(t_{ij})$ must be bounded below whatever the configuration. Schoenberg's theorem [6] gives the second requirement, as for the two-point method above, and so dictates the form of L : written in the Legendre basis, its coefficients in degrees 1 and above must be nonnegative. In this work L is taken to be linear, $L(t) = a_0 + a_1 t$, so the only condition is $a_1 \geq 0$.

For the triangles, subtract a function K of the triangle's three inner products. It must lie below the triangle's share, and $\sum_{i < j < l} K$ must be bounded below whatever the configuration. This is the new ingredient, and the reason the method is called three-point: a pair-based bound never uses the fact that the chords come from points of $\mathbb{R}^3$, whereas the three inner products of a triangle are constrained, since three unit vectors of $\mathbb{R}^3$ cannot meet at arbitrary angles. Bachoc and Vallentin supply the required bound. Their positivity statement is about a matrix rather than a number, so the corresponding free parameter is a positive semidefinite matrix, one for each degree: $K_0, \dots, K_5$ here.

The Bachoc–Vallentin sum runs over *all* ordered triples of points, including degenerate ones in which two of the points coincide. Such a triple has inner products $(1, t, t)$, where t is the inner product of the remaining pair, so its term is a function of that pair alone and belongs with the pair sum. What is subtracted from each pair's share is therefore not $L(t)$ alone but $L(t) + 3K(1, t, t)$, the factor 3 counting the three positions the repeated point can occupy in an ordered triple; the fully degenerate triples, with inner products $(1, 1, 1)$, contribute only a constant, which goes into the bound (§2.6).

A **data structure** packages L , λ , and the matrices $K_0, \dots, K_5$ into one tuple:

$$\mathcal{D} = (a_0, a_1, \lambda, K_0, K_1, K_2, K_3, K_4, K_5). \quad (17)$$

Its entries are as follows.

$a_0, a_1 \in \mathbb{R}$ (**classical**). The coefficients of the **linear potential**

$$L(t) := a_0 + a_1 t. \quad (18)$$

Writing $\mu(x) := \sum_i \sum_j t_{ij} = \|\sum_i x_i\|^2$, the side condition is $a_1 \geq 0$; the resulting **design term** $a_1 \mu(x)$ vanishes at any antiprism, since its eight points always sum to **0** by symmetry.

$\lambda \in \mathbb{R}$, $0 \leq \lambda \leq \frac{1}{18}$. The scalar governing the pair/triangle split in (16), which combines the two separate Bachoc–Vallentin constraints into a single bound. For $N = 8$, taking $\lambda = 0$ gives at best ≈ 19.6730 , short of $E(u^*)$ even with the matrices $K_0, \dots, K_5$, so a positive λ is needed.

$K_0, \dots, K_5$ [2, 3]. The Bachoc–Vallentin matrix variables, six of them as in the example of Cohn and Woo: positive semidefinite matrices of sizes 9, 8, 7, 6, 5, 4 respectively. For $N = 8$, the blocks K_6, K_7, K_8 allowed by the degree-8 truncation are zero in the optimal certificate and are omitted; truncating at degree 4 or 6 instead gives a valid bound that falls short of $E(u^*)$ by 0.023 and 0.006 respectively. With the Bachoc–Vallentin matrices $S_k(u, v, t)$ above, these assemble into the **kernel function**

$$K(u, v, t) := \sum_{k=0}^5 \langle K_k, S_k(u, v, t) \rangle. \quad (19)$$

From here on, $\mathcal{D}$ denotes an arbitrary data structure and $\mathcal{D}^*$ the specific one constructed in §2.8.

2.6 The bound-plus-slack identity

The analogue of the two-point identity (8) has two kinds of slack term instead of one. The two Bachoc–Vallentin constraints are both built from the same three-point kernel K : one evaluated at a repeated triple $(t, t, 1)$, which collapses two of the three points together and so is a statement about pairs; the other at the inner products (u, v, w) of three distinct points, a statement about triangles. In the present notation these give pairSlack and triSlack respectively, their matrix positivity gives BV, and $a_1 \geq 0$ gives the design term:

$$\text{BV}(x) := \sum_{i,j,l=1}^8 K(t_{ij}, t_{il}, t_{jl}), \quad (20)$$

$$\text{pairSlack}(\mathcal{D}, x, i, j) := (1 - 18\lambda) \frac{1}{s_{ij}} - a_0 - a_1 t_{ij} - 3K(1, t_{ij}, t_{ij}), \quad (21)$$

$$\text{triSlack}(\mathcal{D}, x, i, j, l) := \lambda \left(\frac{1}{s_{ij}} + \frac{1}{s_{il}} + \frac{1}{s_{jl}} \right) - K(t_{ij}, t_{il}, t_{jl}), \quad (22)$$

$$\text{design term} := a_1 \mu(x). \quad (23)$$

Adding and subtracting these same quantities in every pairwise and triangle term of the energy, the way $y(s_{ij})$ was added and subtracted for the two-point bound, and summing over the 28 pairs and 56 triangles, collects them into the single quantity

$$2f(\mathcal{D}, x) := \text{BV}(x) + 2 \sum_{i < j} \text{pairSlack}(\mathcal{D}, x, i, j) + 6 \sum_{i < j < l} \text{triSlack}(\mathcal{D}, x, i, j, l) + a_1 \mu(x), \quad (24)$$

the factors 2 and 6 counting each pair and each triangle once for every ordering of its indices, since pairSlack and triSlack depend only on the unordered pair or triangle. With L as above, setting the **bound**

$$b(\mathcal{D}) := \frac{64a_0 - 8L(1) - 8K(1, 1, 1)}{2} = 28a_0 - 4a_1 - 4K(1, 1, 1), \quad (25)$$

to collect the leftover constant terms gives, for every admissible x and every data structure, the exact identity (Bachoc and Vallentin [2] and Cohn and Woo [3] state only the inequality it implies)

$$E(x) = b(\mathcal{D}) + f(\mathcal{D}, x). \quad (26)$$

As with the two-point bound, if $f(\mathcal{D}, x) \geq 0$ then $E(x) \geq b(\mathcal{D})$. The rest of this section constructs a data structure with $b(\mathcal{D}) = E(u^*)$ and $f(\mathcal{D}, x) \geq 0$ for every x satisfying the separation bound (11).

2.7 The polynomials pairP and triP

Two of the four terms are automatically nonnegative for every admissible x : BV from $K_k \succeq 0$, and the design term from $a_1 \geq 0$. The other two, pairSlack and triSlack, need the two Bachoc–Vallentin constraints, specialised to this problem. Clearing denominators in $\text{pairSlack} \geq 0$ and $\text{triSlack} \geq 0$ turns each into a polynomial inequality:

$$\text{pairP}(\mathcal{D}, s) := (1 - 18\lambda) - s \left[a_0 + a_1 \left(1 - \frac{s^2}{2}\right) + 3K\left(1, 1 - \frac{s^2}{2}, 1 - \frac{s^2}{2}\right) \right], \quad (27)$$

$$\text{triP}(\mathcal{D}, p, q, r) := \lambda(qr + pr + pq) - pqr \cdot K\left(1 - \frac{p^2}{2}, 1 - \frac{q^2}{2}, 1 - \frac{r^2}{2}\right), \quad (28)$$

with $\text{pairSlack} = \text{pairP}(\mathcal{D}, s_{ij})/s_{ij}$ and $\text{triSlack} = \text{triP}(\mathcal{D}, p, q, r)/(pqr)$, so $f(\mathcal{D}, x) \geq 0$ follows once $\text{pairP}(\mathcal{D}, \cdot) \geq 0$ and $\text{triP}(\mathcal{D}, \cdot, \cdot, \cdot) \geq 0$ hold on the relevant ranges.

As with the two-point bound, a plain inequality is not enough to make the bound sharp: following Yudin, what is needed is a double zero at the target configuration. At the certificate $\mathcal{D}^*$, $\text{pairP}(\mathcal{D}^*, \cdot)$ has a double zero at each of the four chords A, B, C, D , and $\text{triP}(\mathcal{D}^*, \cdot, \cdot, \cdot)$ vanishes (value and gradient) at each of the five triangle types, pinning $\mathcal{D}^*$'s 24 free entries down to specific numbers, via the linear system built next.

2.8 Constructing $\mathcal{D}^*$: the pivot system and the 24×24 matrix M

The identity (26) holds for every data structure, but for a generic one the resulting bound is not sharp: $b(\mathcal{D}) < E(u^*)$. The data structure $\mathcal{D}^*$ is therefore constructed with $b(\mathcal{D}^*) = E(u^*)$ as one of its defining equations.

The six blocks $K_0, \dots, K_5$ have $45 + 36 + 28 + 21 + 15 + 10 = 155$ independent entries. Two reductions bring the number of unknowns down to 24.

First, equality at the antiprism forces $\text{BV}(\text{antiprism}) = 0$. Summed over the antiprism, $\sum_{i,j,l} S_k(t_{ij}, t_{il}, t_{jl})$ has rank one, say $v_k v_k^\top$, so for $K_k \succeq 0$ this is complementary slackness, $K_k v_k = 0$. Each block is therefore written as $K_k = B_k H'_k B_k^\top$, where the columns of the fixed basis matrix B_k span $v_k^\perp$ (its entries are polynomials in u^* with rational coefficients). This builds $K_k v_k = 0$ in, absorbing 39 linear constraints and leaving $36 + 28 + 21 + 15 + 10 + 6 = 116$ free entries across the H'_k . Together with a_0, a_1, λ that makes 119 numbers.

Second, 95 of these (namely a_0, a_1, λ and 92 entries of the H'_k) are fixed explicit rationals. The remaining 24, the **pivots**, are determined by 24 linear tightness conditions:

rows	condition
1	$b(\mathcal{D}^*) = E(u^*)$
3	$\text{pairP}(\mathcal{D}^*, s_X) = 0$ for $X \in \{B, C, D\}$
4	$\text{pairP}'(\mathcal{D}^*, s_X) = 0$ for all four chords $X \in \{A, B, C, D\}$
5	$\text{triP}(\mathcal{D}^*, \cdot) = 0$ at each of the five triangle types
11	gradient components of $\text{triP}(\mathcal{D}^*, \cdot)$ at the five types (all but one)

Full tightness asks for 26 conditions: the bound, a value and a derivative of pairP at each of the four chords, and a value and the gradient of triP at each of the five triangle types. Since triP is symmetric in its three arguments, a type with a repeated chord has only two distinct gradient components, so the gradients contribute $2 + 3 + 3 + 2 + 2 = 12$ conditions. The 26 conditions have rank 24, and two are dropped. The value $\text{pairP}(\mathcal{D}^*, s_A) = 0$ is dropped because it follows from the

others. Evaluating the identity (26) at the antiprism, where $BV = 0$ (built in via $K_k v_k = 0$ below) and $\mu = 0$, gives the affine relation

$$E(u^*) - b(\mathcal{D}) = \sum_X n_X \frac{\text{pairP}(\mathcal{D}, s_X)}{s_X} + 3 \sum_T m_T \frac{\text{triP}(\mathcal{D}, T)}{p_T q_T r_T}, \quad (29)$$

where X runs over the four chords with counts n_X , and $T = (p_T, q_T, r_T)$ over the five triangle types with multiplicities m_T . Once the bound row and the other eight value rows hold, only the term $n_A \text{pairP}(\mathcal{D}, s_A)/s_A$ remains, and $n_A = 8 \neq 0$ forces it to vanish. One gradient component at the type DCB is dropped because it follows from $E'(u^*) = 0$, by differentiating the identity (26) along the antiprism family. Both dropped conditions are proved in Lean to hold once M below is invertible. The remaining rows sum to $1 + 3 + 4 + 5 + 11 = 24$, exactly the number of pivots, so the system is square. Since $b(\mathcal{D})$, $\text{pairP}(\mathcal{D}, \cdot)$, and $\text{triP}(\mathcal{D}, \cdot)$ are each affine in the pivots once the chord and triangle inputs are fixed, these 24 conditions assemble into a linear system in the pivot vector $\mathbf{p} \in \mathbb{R}^{24}$, and the pivots are *defined* as its solution:

$$M\mathbf{p} = \mathbf{c}, \quad \mathbf{p}^* := M^{-1}\mathbf{c} \quad (30)$$

(`pivotMatrix`, `pivotRhs` and `pivots` in the Lean code). If M is invertible, all 24 tightness conditions hold at $\mathbf{p}^*$ by construction. The 24×24 matrix M and the vector $\mathbf{c} \in \mathbb{R}^{24}$ are built from the fixed entries, the bases B_k , and the chord and triangle data, so, like u^* itself, they are irrational. $\mathcal{D}^*$ is the data structure with the fixed rational entries and the exact pivots $\mathbf{p}^*$.

Why the certificate is implicit. Exact tightness places the pivots in the degree-192 number field generated by $\sqrt{2}$, u^* and the chord lengths, where their coefficients have about 15 000 digits; a data structure with rational entries could prove at best $E(x) \geq E(u^*) - \varepsilon$ for some $\varepsilon > 0$. The warrant therefore keeps $\mathcal{D}^*$ implicit and works with an explicit 22-digit rational vector $\tilde{\mathbf{p}} \in \mathbb{Q}^{24}$ (`pivotsNum` in the Lean code) and a proved enclosure of $\mathbf{p}^*$ around it (fact (i) below). Everything built from $\mathcal{D}^*$ is affine in the pivots, so an inequality checked at $\tilde{\mathbf{p}}$ with a positive margin transfers to $\mathcal{D}^*$, with an explicit error bound. All numerical checking is therefore done in rational interval arithmetic, while the tightness conditions hold exactly.

2.9 The properties of $\mathcal{D}^*$

Definition 2.2 (Realisable triangle region). A triple $(p, q, r) \in [0.9619, 2]^3$ lies in the **realisable triangle region** if, writing $u = 1 - \frac{p^2}{2}$, $v = 1 - \frac{q^2}{2}$, $w = 1 - \frac{r^2}{2}$ for the corresponding inner products, the Gram matrix $\begin{pmatrix} 1 & u & v \\ u & 1 & w \\ v & w & 1 \end{pmatrix}$ is positive semidefinite, equivalently

$$1 + 2uvw - u^2 - v^2 - w^2 \geq 0. \quad (31)$$

This is exactly the condition for (p, q, r) to be realisable as the three chord lengths of an actual spherical triangle $\{x_i, x_j, x_l\} \subset S^2$.

The warrant needs four facts about $\mathcal{D}^*$. Fact (ii) gives $BV(x) \geq 0$; facts (iii) and (iv) give $\text{pairP}(\mathcal{D}^*, \cdot) \geq 0$ and $\text{triP}(\mathcal{D}^*, \cdot, \cdot, \cdot) \geq 0$; fact (i) makes $\mathcal{D}^*$ well defined and allows the other three to be checked at the rational vector $\tilde{\mathbf{p}}$ instead of the irrational $\mathbf{p}^*$.

- (i) The matrix M is invertible, and $\|\mathbf{p}^* - \tilde{\mathbf{p}}\|_\infty \leq 2 \times 10^{-21}$. (Facts (ii)–(iv) need only 10^{-12} .)
- (ii) Each block H'_k , and hence each $K_k = B_k H'_k B_k^\top$, is positive semidefinite.
- (iii) We have $\text{pairP}(\mathcal{D}^*, \cdot) \geq 0$ on $[0.9619, 2]$.
- (iv) We have $\text{triP}(\mathcal{D}^*, \cdot, \cdot, \cdot) \geq 0$ on the realisable triangle region.

2.10 Verifying the properties using the Lean kernel

Facts (i)–(iv) are proved by reflection. For each fact, a Lean checker evaluates the relevant polynomials in fixed-point interval arithmetic on rational data (the fixed entries, $\tilde{\mathbf{p}}$, and rational enclosures of u^* and the chord lengths), and a soundness theorem transfers its output to the real statement. The checkers are evaluated in the kernel with `decide +kernel`, so the warrant uses only Lean’s three standard axioms; the largest check, part of fact (iv), takes about 30 CPU-hours.

Facts (ii)–(iv) are checked at $\tilde{\mathbf{p}}$ with a positive margin. Since everything is affine in the pivots, fact (i) bounds the change in moving to $\mathbf{p}^*$ by 10^{-12} times an explicit rational ℓ^1 -norm, which the margin absorbs. Facts (iii) and (iv) are checked by subdividing their domains into boxes. At the four chords and five triangle types, where `pairP` and `triP` vanish, a certified quadratic lower bound is used instead, and the perturbation bound is applied only after these zeros have been divided out.

The fixed entries and $\tilde{\mathbf{p}}$ come from a floating-point semidefinite-programming solve outside Lean.

2.11 Uniqueness

Combining the identity with facts (i)–(iv) shows $E(x) \geq E(u^*)$ for every admissible x , with equality at the antiprism itself. This subsection sketches why the antiprism is also the *only* minimiser, up to the symmetries of the sphere and of the point labels.

At a configuration x attaining $E(x) = E(u^*)$, the identity forces $f(\mathcal{D}^*, x) = 0$; since $f(\mathcal{D}^*, x)$ is a sum of individually nonnegative terms, each one (`BV(x)`, every `pairSlack`, and every `triSlack`) must vanish separately. Strict versions of the box-covering and boundary arguments show that `pairP`($\mathcal{D}^*, \cdot$) is positive except at the four double zeros, so every chord of x is one of A, B, C, D ; the same certified margin forces every triangle to be one of the five types.

A finite search over ways of assigning these chord types to the edges between the eight points, consistent with the triangle constraints, narrows the possibilities down to relabellings of the antiprism’s own assignment, together with one further pattern that passes the triangle-type check: an **aligned quadruple**, four points split into two disjoint pairs of chord type A with all four remaining cross-pairs of chord type B , so that every one of its four triangles is the valid type BBA . This is excluded separately: the 4×4 Gram matrix such a quadruple would require has a provably nonzero determinant, yet any four points of $\mathbb{R}^3$ are linearly dependent and so must have a singular Gram matrix, a contradiction, so no aligned quadruple can occur among points of S^2 . Matching Gram matrices then finish the argument via a classical isometry lemma, giving x as a rotated, relabelled copy of the antiprism.

3 Conclusion

The work done in Section 2 provides the arguments that our conclusion relies on. There are two cases. If an admissible x has $E(x) > E(u^*)$, there is nothing to prove. Otherwise $E(x) \leq E(u^*) = 19.675288\dots \leq 19.6753$, so the separation theorem applies, every chord of x lies in $[0.9619, 2]$, and the four facts give $f(\mathcal{D}^*, x) \geq 0$; the identity then yields

$$E(x) = b(\mathcal{D}^*) + f(\mathcal{D}^*, x) \geq b(\mathcal{D}^*) = E(u^*). \quad (32)$$

Either way, $E(x) \geq E(u^*)$ for every admissible x . The bound is attained by the optimal antiprism. So the optimal square antiprism minimises the Coulomb energy among *all* 8-point configurations on S^2 , and, as Section 2.11 shows, is the *only* minimiser up to rotation of the sphere and relabelling of the points.

References

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